

Technical Requirements Document (TRD)

Project: The Zaika Digital Ordering Ecosystem

Version 1.0

Product Type: Direct Ordering Platform + Loyalty Ecosystem + Restaurant Growth Engine

1. Technical Vision

The Zaika platform should not be built as a traditional restaurant website.

It should be engineered as:

Food Commerce Platform

Similar architecture philosophy:

- DoorDash Storefront
- Sweetgreen
- CAVA
- Starbucks Ordering
- Behrouz Ordering

while remaining lightweight enough for a single-location business.

2. Technical Objectives

Primary Objectives

Performance

- Lighthouse Score > 95
- Mobile Performance > 90

- Core Web Vitals Pass
-

SEO

- Rank for local searches
 - Google Rich Snippets
 - Structured Data
-

Conversion

Target:

- Menu View → Cart > 15%
 - Cart → Checkout > 60%
 - Checkout → Order > 85%
-

Scalability

Support:

Phase 1

1 Location

↓

Phase 2

5 Locations

↓

Phase 3

50+ Franchise Locations

without major rewrite.

3. System Architecture

Customer

↓
Next.js Frontend
↓
API Layer
↓
Supabase Backend
↓
Database

Additional Services

Razorpay
WhatsApp
Google Maps
Analytics
Email
SMS

4. Technology Stack

Frontend

Framework

Next.js 15+

Reason:

- SEO
 - SSR
 - Fast
 - Scalable
-

Language

TypeScript

Reason:

- Type safety
- Maintainability

Styling

TailwindCSS

Reason:

- Faster development
- Design consistency

UI Components

Shadcn/UI

Reason:

- Premium feel
- Accessibility

Animations

Framer Motion

Reason:

- Smooth interactions

5. Backend

Backend Platform

Supabase

Modules:

- PostgreSQL
- Authentication
- Storage
- Realtime
- Edge Functions

6. Database Design

Users Table

users

Fields:

Field	Type
id	UUID
name	Text
phone	Text
email	Text
created_at	Timestamp
last_login	Timestamp

Addresses Table

addresses

Field	Type
id	UUID
user_id	UUID
addresses	Text
city	Text
pincode	Text
lat	Float
lng	Float

Categories Table

categories

Examples:

- Akni
- Biryani
- Mughlai
- Kathiyawadi

Menu Items Table

menu_items

Field	Type
id	UUID
name	Text
description	Text
image	Text
price	Decimal
category_id	UUID
active	Boolean

Orders Table

orders

Field	Type
id	UUID
user_id	UUID
status	Enum
total	Decimal

payment_status Enum

created_at Timestamp

Order Items Table

order_items

Stores purchased products.

Loyalty Table

loyalty_points

Stores:

- earned
 - redeemed
 - balance
-

Coupons Table

coupons

Stores:

- code
 - expiry
 - discount
-

Referral Table

referrals

Stores:

- referrer

- referred
 - rewards
-

7. Authentication

Login Options

Phase 1

OTP Login

via mobile number

Phase 2

Google Login

Phase 3

Apple Login

Authentication Flow

Phone Number



OTP



Verify



Create User



Dashboard

8. Menu Management System

Requirements

Admin should:

- Add item
 - Edit item
 - Delete item
 - Mark unavailable
 - Upload image
-

Categories

Admin can:

Create

Update

Delete

Sort

Dynamic Pricing

Support:

- Festival pricing
 - Happy hour pricing
 - Promotional pricing
-

9. Cart System

Requirements

User can:

- Add item
 - Remove item
 - Increase quantity
 - Decrease quantity
-

Cart Persistence

Guest Cart

↓

Login

↓

Merge Cart

Storage

Local Storage

-

Database Sync

10. Checkout System

Step 1

Address

Step 2

Delivery Slot

Step 3

Payment

Step 4

Confirmation

Maximum:

4 screens

11. Payment System

Gateway

Razorpay

Supported

- UPI
 - Credit Card
 - Debit Card
 - Net Banking
 - Wallet
-

Webhooks

Required

payment_success
payment_failed
refund_processed

12. Order Management

Status Workflow

Pending



Accepted



Preparing



Out for Delivery



Delivered

Alternative

Cancelled

Real-time Updates

Customer sees:

Live status updates

without refresh.

13. Delivery System

Phase 1

Manual Delivery

Admin assigns:

- Rider
- Delivery estimate

Phase 2

Third-party Integration

Potential:

- Shadowfax
 - Porter
 - Dunzo alternatives
-

14. Loyalty Engine

Rules

₹100 spent

↓

10 Points Earned

Redemption

100 Points

↓

₹10 Discount

Configurable from Admin.

15. Referral Engine

Flow

User gets referral code.

Friend orders.

Both rewarded.

Rules configurable.

16. Wallet System

Stores:

- Refunds
 - Referral rewards
 - Loyalty rewards
-

Fields

wallet_balance
wallet_transactions

17. Notification System

Channels

WhatsApp

Email

SMS

Push Notifications

Events

Order Placed

Order Accepted

Order Ready

Order Delivered

Reward Earned

18. WhatsApp Integration

Critical Feature

Provider

Meta API

or

Interakt

Messages

Order Updates

Offers

Rewards

Feedback Requests

19. Review System

Internal Reviews

Customer can rate:

1–5 stars

Review Categories

Food

Packaging

Delivery

Experience

Auto Request

After delivery:

Rate Your Order

20. Search System

User can search:

- Akni
 - Biryani
 - Chicken
 - Kebabs
-

Search Requirements

Response:

< 100ms

21. Personalization Engine

Show:

Previously Ordered

Recommended For You

Popular Near You

Future:

AI Recommendations

22. Admin Dashboard

Dashboard Widgets

Today's Orders

Revenue

Top Selling Item

Conversion Rate

Returning Customers

Modules

Orders

Menu

Customers

Offers

Analytics

Reviews

Settings

23. Analytics System

Customer Analytics

Track:

- Acquisition Source
- First Order
- Last Order
- Lifetime Value

Product Analytics

Track:

- Most ordered
- Least ordered
- Highest profit

Funnel Analytics

Track:

Landing

↓

Menu

↓

Cart

↓

Checkout

↓

Purchase

24. SEO Requirements

Technical SEO

- Sitemap XML
 - Robots.txt
 - Canonicals
 - Open Graph
-

Structured Data

Restaurant Schema

Menu Schema

Review Schema

FAQ Schema

Local SEO

Google Business Profile integration.

25. Security Requirements

Authentication

JWT

Passwords

Hashed

Payment

PCI compliant via Razorpay

API Security

Rate limiting

CSRF protection

Input validation

26. Performance Requirements

First Load

< 2 seconds

Mobile

< 3 seconds

API Response

< 300ms

Image Optimization

WebP

AVIF

Responsive Images

27. Accessibility Requirements

WCAG 2.2 AA

Requirements:

- Keyboard navigation
 - Screen readers
 - Contrast compliance
 - Focus states
-

28. Infrastructure

Hosting

Frontend:

[Vercel](#)

Backend:

[Supabase](#)

Storage:

Supabase Storage

CDN:

Global Edge Network

29. Future Roadmap Architecture

Phase 1

Marketing Website

-

Direct Ordering

Phase 2

Loyalty

-

Referral

-

CRM

Phase 3

Progressive Web App (PWA)

Offline support

Home screen install

Push notifications

Phase 4

Native Apps

Android

iOS

Phase 5

Multi-Branch Architecture

Support:

- Ahmedabad
- Gandhinagar
- Vadodara
- Surat

from a single admin panel.

30. Recommended Enterprise Additions (Important)

To truly compete with aggregator platforms, add these modules from Day 1:

Customer Data Platform (CDP)

Single customer profile.

Marketing Automation

- Abandoned cart recovery
 - Win-back campaigns
 - Birthday offers
 - Repeat-order reminders
-

Smart Offers Engine

Example:

Customer ordered Akni 3 times

Automatically offer:

"Get 20% off on Family Akni Combo"

QR Loyalty Program

Every package should contain:

- QR Code
- Referral Link
- Loyalty Enrollment Link

This converts aggregator customers into direct customers over time.

Final Technical Principle

The platform should be architected as a **commerce and retention system**, not as a restaurant website.

The website UI is only the front layer.

The real business value comes from:

Customer Data + Loyalty + Referrals + Direct Ordering + Automation

Those five systems will generate far more long-term profit than menu pages, galleries, or restaurant information sections.